

IB MATH AA SL

Lesson 26

Standard Deviation & the GDC

Statistics & Probability · 60-minute lesson

Syllabus SL 4.3 · Property of Lynda Badmus Education

A large, stylized sigma symbol (σ) is positioned in the upper right quadrant of the slide. It is rendered in a light blue color against a dark blue background. The symbol is partially enclosed by a large, dark blue circular shape that overlaps the right edge of the slide.

Do Now — recall

ON YOUR WHITEBOARD

1. Which is more spread out: $\{5,5,5\}$ or $\{1,5,9\}$?
2. What does 'spread' mean for data?
3. Mean of 2, 4, 6?

Answers (click to reveal)

1. $\{1,5,9\}$
2. how far values are from the centre
3. 4

What we're learning today

LEARNING OBJECTIVES

- 1 Understand**
standard deviation as a measure of spread.
- 2 Use**
the GDC to find mean and σ .
- 3 Compare**
data sets using mean and σ .
- 4 Interpret**
what a large or small σ means.

SUCCESS CRITERIA — I CAN...

- ✓
explain what standard deviation measures
- ✓
enter data into the GDC's stats menu
- ✓
read off the mean and standard deviation
- ✓
compare two data sets fairly

Standard deviation

SPREAD ABOUT THE MEAN

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Small σ

data clustered near the mean

Large σ

data widely spread

In the exam

you find σ with the GDC, not by hand

GDC STEPS

1.
enter data into a list
2.
1-Variable Statistics
3.
read $\bar{x}$ and σ_x (the population SD)

Read off mean & σ

QUESTION

Using a GDC, the data 4, 7, 7, 9, 13 gives mean 8 and $\sigma \approx 2.97$.

What does σ tell us?

SOLUTION

Mean $\bar{X} = 8$ (the centre).

$\sigma \approx 2.97 \rightarrow$ on average values lie about 3 units from the mean.

A small σ would mean the data is tightly packed.

Your turn — interpret

ANSWER IN YOUR BOOK

1. Set A: $\sigma = 1.2$; Set B: $\sigma = 8$.

Which is more consistent?

2. If every value is the same, what is σ ?
3. Does adding 5 to every value change σ ?

Answers (click to reveal)

1. Set A (smaller σ)
2. 0
3. No — spread is unchanged

Comparing data sets

MEAN + SPREAD

compare $\bar{x}$ and σ

Mean

compares the typical/centre value

σ

compares consistency / spread

Always

comment on BOTH when comparing

EXAMPLE WORDING

'Class A has a higher mean, so scored better on average,'

'but a larger σ , so was less consistent.'

Compare two classes

QUESTION

Class A: mean 62, $\sigma = 4$.

Class B: mean 62, $\sigma = 11$.

Compare the two classes.

SOLUTION

Same mean $\rightarrow$ same average score.

Class A has the smaller σ ($4 < 11$).

So Class A is more consistent; B's results vary much more.

Exam-style question

The masses (kg) of 6 parcels are 2, 3, 3, 5, 6, 11.

- (a) Use your GDC to find the mean. [2]
- (b) Find the standard deviation. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) mean = $30/6 = 5$ (M1)(A1)
- (b) $\sigma \approx 2.94$ (from GDC, 3 s.f.) (M1)(A1)

Common mistakes

✗ **σ measures spread**
not the centre — that's the mean.

✗ **Use the GDC**
Don't compute σ by hand under exam time.

✗ **Compare both**
State mean AND σ when comparing data sets.

✗ **Shift invariance**
Adding a constant changes the mean, not σ .

Mixed practice

QUESTIONS

1. Which σ means more spread: 2 or 9?
2. Mean of 3, 3, 3, 3.
3. σ of 5, 5, 5, 5.
4. Higher mean but higher σ — what does it mean?

Answers (click to reveal)

1. 9
2. 3
3. 0
4. better on average but less consistent

Plenary & exit ticket

KEY FORMULAS

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

small $\sigma \Rightarrow$ consistent

EXIT TICKET — before you leave

1. Does σ measure centre or spread?
2. σ when all values equal?
3. Bigger σ = more or less consistent?

Answers (click to reveal)

1. spread
2. 0
3. less consistent

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Standard deviation.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 27

Correlation & linear regression — lines of best fit.

$$y = ax + b$$